Benjamin Tran-Pugh

647-768-1669 | btranpug@uwaterloo.ca | linkedin.com/in/benjamin-tran-pugh | github.com/BenjaminTP | benjamintranpugh.com

EDUCATION

University of Waterloo **Sep 2025 – Jun 2030**
Candidate for BASc, Mechanical Engineering (Co-op) *Waterloo, ON*
Cumulative GPA: 3.83/4.0

TECHNICAL SKILLS

CAD & Analysis: SolidWorks (FEA, PDM, Sheet Metal), Onshape (FEA), Fusion 360, Autodesk Inventor, AutoCAD, GD&T
Software: Python (NumPy, SciPy, Matplotlib), C/C++ (ESP32), MATLAB, Linux (Raspberry Pi, Jetson Orin Nano), Git
Other: 3D Printing (PLA, PETG, TPU, ABS, PA6-CF), DFM/DFA, Machining, Technical Drawings, Microsoft Suite, URDF

EXPERIENCE

Mechanical Engineering Intern **Feb 2026 – Aug 2026**
Human Computer Lab (SPC '25) *Toronto, ON*

- Owned mechanical and actuation development of LeLamp, an animatronic robotics product, across three actuator generations, advancing it from functional prototype toward manufacturing readiness through EVT/DVT stages.
- Designed and hardware-validated a spring-based gravity-compensation system to eliminate actuator thermal overload, deriving joint balance conditions from Lagrangian mechanics and verifying against published literature
- Designed mechanisms such as antagonistic pairs, capstan drive, and full/partial spring gravity compensation by running through calculations, simulating physics, testing real-world results against simulations, and iterating
- Led actuator trade studies using Pugh matrices with explicit kill criteria; recommendations adopted into the product
- Identified reliability gaps in fatigue and lifecycle coverage and drove their integration into the validation roadmap from program start, preventing product failure, safety issues, and required redesign down the line
- Identified root causes of motor noise from first principles and reduced most vibrational resonance of adjacent walls

Mechanical Team Member **Sep 2025 – Present**
Waterloo Aerial Robotics Group Waterloo, ON

- Designed and fabricated a sub-60 g payload-release mechanism for the Valkyrie competition drone, securing objects of varying size and mass in flight and releasing them at a small target with snag-free, propeller-safe routing
- Designed iterations of empennage structures for Helios, a fixed-wing aircraft evolving toward VTOL capability
- Managed design data in SolidWorks PDM across a multi-hundred-member team, integrating mechanical designs with electrical and autonomy subsystems, to help team win the Pip Rudkin Perseverance Award at AEAC 2026

PROJECTS

Double Pendulum Chaos Simulation **Python, SciPy, NumPy**

- Derived the full nonlinear equations of motion from first principles by hand from the Euler-Lagrange equation, obtaining closed-form angular accelerations via Cramer's rule, which allowed physics simulations to run
- Built a numerical simulation using python (NumPy, Matplotlib, and SciPy) demonstrating sensitive dependence and chaotic motion arising from variations in initial conditions as small as 1e-9 rad
- Computed and visualized a time-to-flip fractal across the initial-condition space using vectorized NumPy

4-DOF Robotic Arm 2.0 **SolidWorks, ESP32, C++**

- Modeled a 50+ part assembly integrating off-the-shelf components; validated 8 structural components with FEA against load requirements, ensuring a factor of safety of 1.5
- Implemented 4-DOF inverse kinematics, trajectory smoothing, and a web-based control interface using C/C++ running directly on the ESP32 for the built-in WiFi and Bluetooth capabilities
- Wrote Python and MATLAB simulations to verify joint limits and workspace before hardware runs, preventing clashing, tipping, and overheating

Pneumatic Launcher **Fusion 360**

- Designed a pneumatic launch system achieving 60+ m range on a $100 budget; sized the pressure chamber to a factor of safety of 2 on burst pressure